

Data-driven *pricing*

Starting with luxury event venues — where every date is a one-off and every price is a negotiation.

01 THE PARTNER

A Manhattan event venue selling event days at

\$50K–\$200K

Galas, conferences, celebrations — booked one date at a time.

02 THE INCUMBENT

A static rate card

Varies only by year, month and weekday. It never sees:

- Event type
- Lead time
- Demand pressure

Every quote is negotiated by hand — and the hardest call, *take this deal or hold the date for a better one*, is made on gut feel.

03 MLIQ

Replace the card with models trained on the venue's own sales history.

Delivered inside the existing Salesforce workflow — pricing and lead scoring where the sellers already work.

Predict, then *prescribe*

Stage 1 predicts what shows up and what it's worth. Stage 2 prescribes the call.

PREDICTIVE · MODEL 1 — DEMAND

What kind of demand does this date attract?

Event type and calendar footprint. A conference books midweek and consumes 2–4 days with load-ins; a gala takes one.

PREDICTIVE · MODEL 2 — VALUE

What should this date cost?

By event type, section, weekday and month — as a **negotiable range**, not a single number.

PRESCRIPTIVE — PRICING THE OPPORTUNITY COST

Accepting a one-day social on a Wednesday destroys the **Wed+Thu** and **Tue+Wed** conference blocks — the two most common multi-day shapes in the book.

\$195K

MEDIAN BLOCK VALUE

ONE DECISION RULE

Hold the date if $P(\text{better arrives}) \times E[\text{value}] - \text{holding cost} > \text{the offer in hand}$

THE BAR TO CLEAR — MAE, LOWER IS BETTER

56.4% HEAD-TO-HEAD VS THE RATE CARD



Curves fitted to realized bookings already beat the card — *in-sample today*. The bake-off re-measures it on an out-of-time holdout. **The models must clear the fitted curves, not the card.**

One venue's history, *2021–2026*

Extracted and anonymized end-to-end: pseudonyms, an allow list of approved fields, and no names, emails or free text leaving the CRM.

3,985

OPPORTUNITIES – THE DEALS

5,877

RESERVATIONS – THE CALENDAR

1,170

LOST OPPORTUNITIES RESTORED

\$131M

QUOTED-AND-LOST REVENUE RECOVERED IN CLEANUP

THE HOLE

2021–2023 was migrated from a legacy CRM *as wins only*.
The lost deals existed as empty shells — or not at all.

266

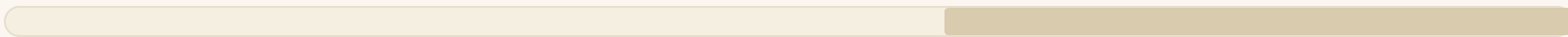
REBUILT FROM SCRATCH

904

EMPTY SHELLS REPAIRED

USABLE DEMAND HISTORY

BEFORE – AS DELIVERED



AFTER – RECONSTRUCTED



2021 2022 2023 2024 2025 2026

Demand history now runs from 2022, not 2024. The modeling window doubled.

Finding the *grain*

The unit of analysis had to be earned — the calendar lies about money.

THE TRAP

Reservation money fields are formulas that read the **parent Opportunity**. Summing day money off the calendar inflates revenue

2.5×

THE FIX

Rebuild per-day dollars from the **product lines underneath** — they reconcile to opportunity totals to the cent.

Reconciled to the cent

THE TARGET

Venue rent per event day

\$80,000

MEDIAN WON · FULL VENUE \$90K · PARTIAL \$34.75K

MODELING GRAIN 1

3,706

EVENTS — ONE PER INQUIRY: OUTCOME, EVENT TYPE, TRUE INQUIRY AND LOSS DATES

MODELING GRAIN 2

5,851

EVENT-DAYS — ONE PER DAY PER VENUE SECTION: ITS OWN HOLD POSITION, ITS OWN DOLLARS

OUTCOMES

659 2,981 66

WON LOST OPEN



4.5:1 lost to won — exactly why a model that learns only from wins **misreads the market**.

Inquiry to *Booked*

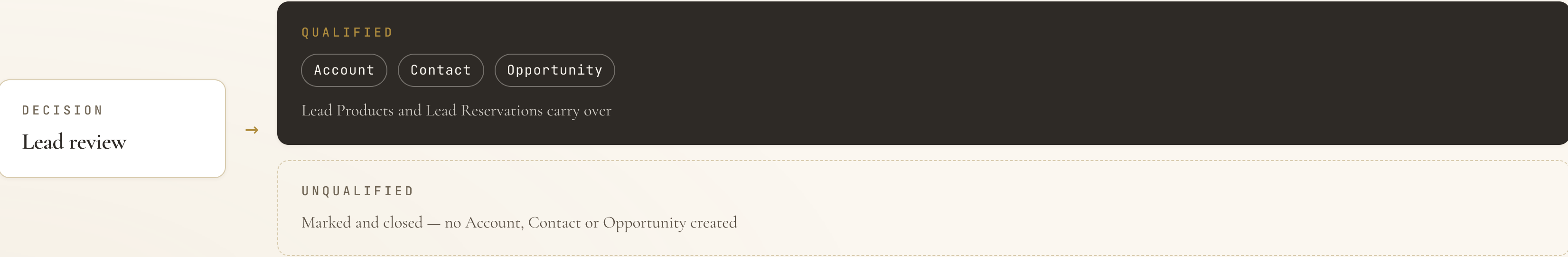
Every inquiry lands as a Lead. Once qualified it becomes an Opportunity that holds dates on the master calendar.

01 CAPTURE — ANY CHANNEL, ONE ENTRY POINT



Each Lead Reservation looks up **one** Lead Product — and the Lead.

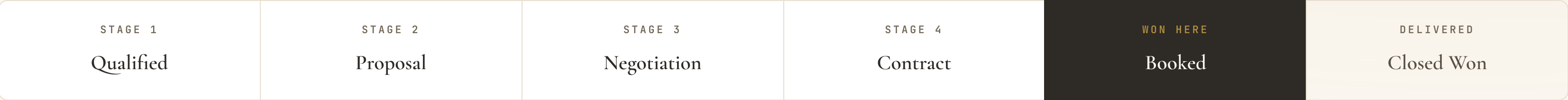
02 QUALIFY — SALES REVIEWS EVERY LEAD



Pipeline, holds and *money*

The money rides on the products attached to each day.

03 PIPELINE — OPPORTUNITY STAGES



Won at Booked. Contract signed, deposit in. Closed Won is bookkeeping after delivery and deposit refund.

Closed Lost. Exit from any stage — mostly on *date availability* and *price*.

04 CALENDAR AND MONEY

MASTER CALENDAR

Reservation

Each product on the Opportunity — every day, load-in included

CHILD OF OPPORTUNITY

Opportunity Product

Venue rent · Beverage · Staffing — one row per item, per day

Per-day pricing. Money sits on each day's line items, not the Opportunity as a whole.

HOLD POSITION — ONE DATE, MANY DEALS

1st First claim on the date

2nd Moves up if the 1st is released

3rd Still tracked and quotable

WIN

Holds flip to **Confirmed**; date blocked.

LOSS

Date released; **next hold moves up**.

The two *predictive* models

Different grains, different training sets — one decision.

	MODEL 1	MODEL 2
	Demand	Value
ESTIMATES	Arrival rate λ per demand shape, by weekday, month and days-out. P(better deal in k days)	Venue rent per event day, as a low / point / high range — every price here is negotiated.
ALGORITHM	Poisson arrival model, partial pooling across sparse shape cells	Gradient-boosted quantile regression on log rent/day
GRAIN	Event (3,706) — an event arrives once, then occupies days; day grain would count a 3-day conference three times	Event-day (5,851) — the same event's Thursday and Saturday are worth different money
FEATURES	weekday, month, lead time, demand shape (type × footprint), market segment	event type, venue section, weekday, month, guest count, lead time, footprint length, segment, industry
TRAINS ON	All arrivals, won AND lost — a deal lost to a blocked date is displaced demand, not a price rejection	Priced won days + clean rejections only; blocked-date losses excluded — their high quotes would teach “ <i>high prices lose</i> ”

THE DEMAND SHAPES ARE REAL CELLS, NOT THEORY

1-day conference-family		973
1-day gala		583
1-day reception		452
1-day wedding		362
2-day conference		244
3-day conference		73

Thin tails share strength through partial pooling; the **multi-day corporate shapes** are the \$195K blocks the opportunity-cost math protects.

ONE SHARED TRAIN / SERVE CONTRACT

9 required + 5 optional features — leakage forbidden by construction: rules-engine outputs, post-outcome fields and dead source fields **raise an error** if they reach a model.

Alternatives

Every candidate is judged on the same rows, the same splits, once.

ALTERNATIVE	GRAIN	VERDICT
Multinomial event-type model Advisor's suggestion: $P(\text{event type} \mid \text{weekday, month, year})$	Event	ADOPTED As the demand-shape baseline inside Model 1 — but arrival intensity stays separate: <i>what</i> books Wednesdays and <i>how often</i> anything arrives move independently.
Summed one-day opportunity costs Price a hold as the sum of individual day costs	Event-day	KEPT AS LOWER BOUND It undercounts block displacement — a Wednesday booking kills every multi-day block crossing Wednesday — so destroyed blocks are priced jointly at event grain.
Neural networks Feed-forward net for value; neural point-process for arrivals	Same as the champion each challenges	RUN AS CHALLENGERS Viable now the reconstruction doubled the history — identical rows and partitions, promoted only on shared-holdout wins.

ONE PARTITION DESIGN — OUT-OF-TIME 60 / 20 / 20



Ordered by event date, cut at event boundaries, grouped by account — **no client and no multi-day event straddles a split**. The final 20% is one shared holdout, **scored once**.

WHY GBM STARTS CHAMPION

On ~3,700 tabular events: native quantile loss for negotiated ranges, monotonic constraints, and attributions a revenue manager can audit.

Whatever wins must also beat the re-fit curves on the holdout. **Nothing ships on architecture appeal.**

What we're claiming — and what would *prove us wrong*

The data work is done. What's left is the measurement we've already committed to.

01 THE CLAIM

A model trained on this venue's own history prices a date better than a card that cannot see event type, lead time or demand pressure.

And it prices the harder question the card never asks — *hold, or take the offer in hand?*

OPEN — 01

EDIT ME: the first thing you want the room to weigh in on.

02 THE PROOF

One out-of-time holdout, scored once.

Champions must beat the **re-fit curves** — not the rate card. Beating the card is the floor, not the result.

No leakage by construction; nothing promoted on architecture appeal.

OPEN — 02

EDIT ME: the second open question.

03 THE FAILURE WE ALREADY PRICED

A model that learns only from wins.

4.5:1 lost to won is why the 1,170 lost opportunities were rebuilt *before* anything was fit.

A blocked-date loss is displaced demand — not a price rejection. The two models split on exactly that line.

OPEN — 03

EDIT ME: the ask — what you need from them before the next checkpoint.